Caught in the Algorithm

Predicting Academic Performance from Social Media Addiction Patterns

BANA 273 – Machine Learning for Analytics | Fall 2025

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Why This Matters

How do social media addiction patterns and usage behaviors relate to students' self-reported academic impact?

Rising Screen Time

Students now spend an all-time high on social media, averaging ≈ 4.5 hours per day.

Sleep Deprivation

93% of Gen Z report staying up past their bedtime due to social media use.

Mental Health Concerns

Studies links heavy or problematic social media use to higher rates of depression and anxiety among young people.

Academic Vulnerability

These behavioral patterns increase the risk of academic difficulties and lower performance.

Data Overview

All data is self-reported and may be subject to response bias.

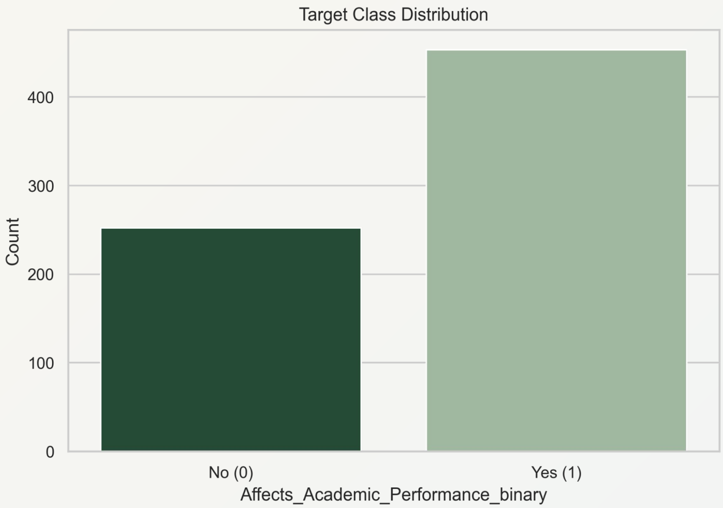
→ Kaggle dataset with 705 survey responses from students aged 16-25

→ Self-reported responses on social media usage, addiction symptoms, conflicts, sleep, and mental health

→ Includes students from 100+ countries

→ **Target variable:**
Affects_Academic_Performance (Yes/No)

→ **Key predictors:**
daily usage hours, addiction score, sleep hours, mental health score, and conflicts over social media



Data Quality & Preprocessing

Data Quality Checks

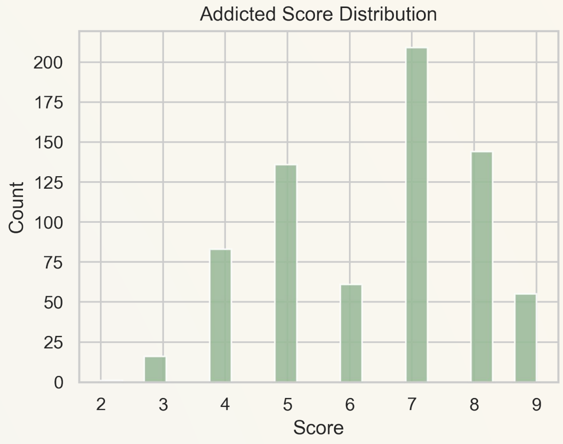
→ Checked dataset for missing values (none found)
→ Verified clean numeric ranges

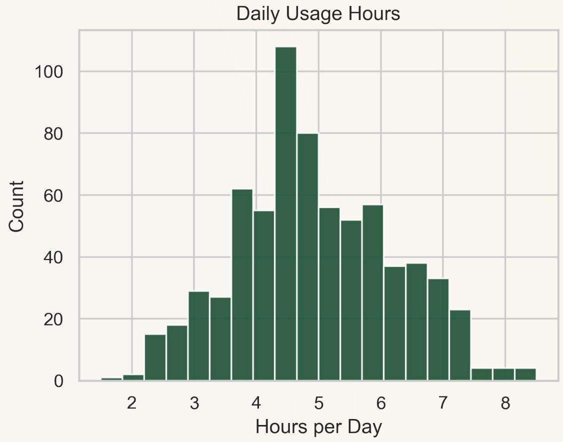
Preprocessing

→ Standardized numeric features for stable coefficient estimation
→ Encoded categorical variables

Modeling Preparation

→ Applied *stratified 10-fold cross-validation* to reduce variance due to the small sample size.





Exploratory Data Analysis

Highly Correlated Predictors

Variables such as Addicted Score and Conflicts over Social media show strong correlations with academic impact.

Clear Separability

These strong linear relationships indicate that the two classes are easily separated.

Well-Being Patterns

Lower sleep hours and lower mental health scores associate with higher self-reported academic impact.

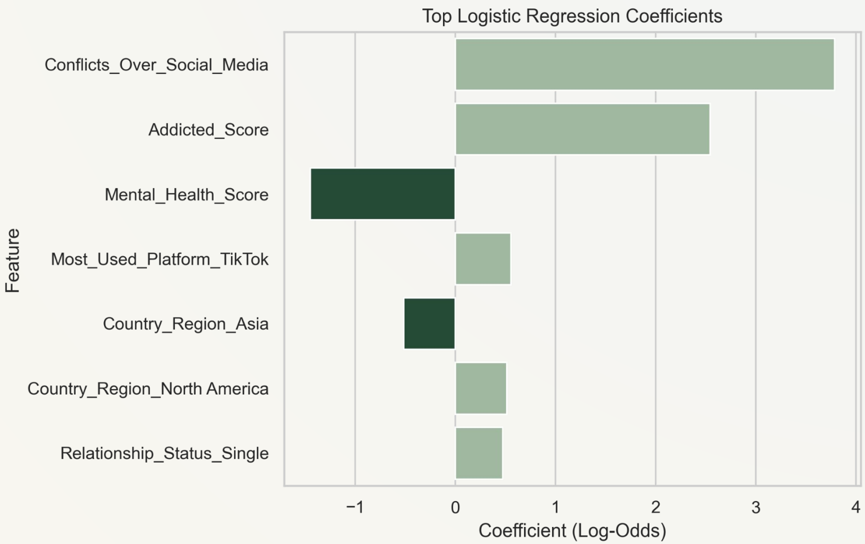
Logistic Regression

Model 1

→ Baseline model to identify key predictors of academic performance.

→ **Reached 100% accuracy with 10-Fold Cross Validation**

Effects on Academic Performance	
Strong positive effects	Strong negative effects
• Addiction • Conflicts	• Sleep • Mental Health



Random Forest

Model 2

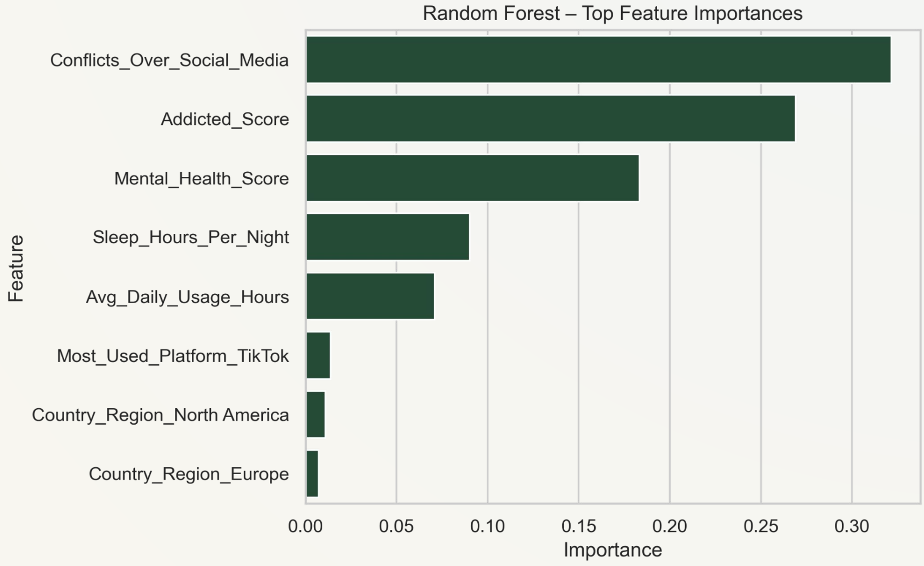
→ Ability to confirm key predictors through stable feature importance

Top predictors (consistent across folds):

- Conflicts Over Social Media
- Addiction Score
- Mental Health Score
- Sleep Hours
- Daily Usage Hours

→ Demographics & platform choice had minimal or no contribution

→ **Confirms academic impact is driven by behavioral patterns, not identity**



Performance Summary

→ Identical Performance Across Models

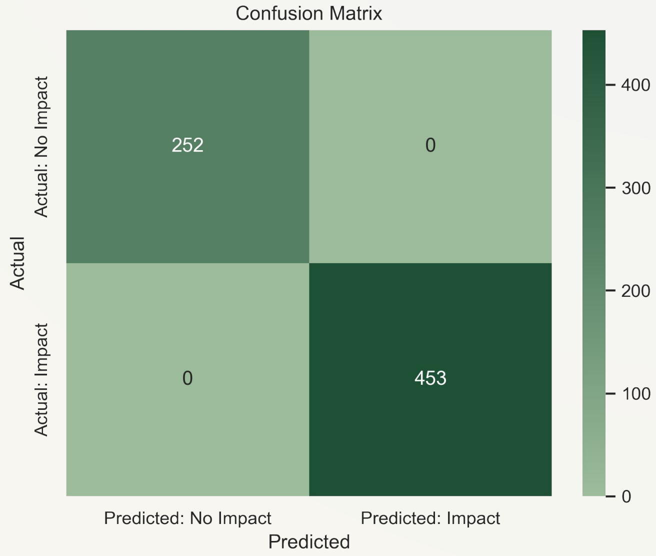
Both models achieved perfect accuracy, precision, recall, and F1 under stratified 10-fold cross-validation

→ No Signs of Overfitting

Robustness across all folds demonstrates that the findings are stable rather than evidence of overfitting

→ Reinforced Behavioral Drivers

Top features remain consistent across approaches.



*Insights**

**Strong patterns reflect dataset structure - to be explained in "Limitations"*

Addction Score is the strongest driver.
Higher compulsive use → sharply higher likelihood of academic disruption.

Social Media Conflicts intensify academic strain
It is particularly important when paired with addictive behavior.

Higher Daily Usage Hours aligns with higher academic disruption

Conflicts on Social Media signal Academic Struggle.

Less Sleep → Poorer Academic Performance

Lower Mental Health Scores amplify effects of addiction and conflicts.

Implications

- **Early Identification of At-Risk Students**
  Behavioral patterns can help flag students who may need academic or mental-health support.
- **Targeted Digital Wellness Programs**
  Interventions promoting healthier usage habits may help reduce academic disruption.
- **Mental Health & Sleep Awareness Initiatives**
  Encourages campaigns that help students build healthier routines around rest and mental health.
- **Conflict Management & Online Boundaries**
  Promotes guidance on managing social-media conflicts and setting digital limits.

Limitations

Implications reflect associations, not causal effects.

→ **Highly Separable Dataset**
Both models achieved perfect accuracy, precision, recall, and F1 under stratified 10-fold cross-validation

→ **Self-Reported Survey Data**
Robustness across all folds demonstrates that the findings are stable rather than evidence of overfitting

→ **Cross-Sectional Data**
Single-time-point data captures associations rather than causal effects or changes in behavior over time

→ **Limited Contextual and Academic Detail**
Demographic categories are broad, and the dataset lacks objective academic metrics , making it difficult to contextualize self-reported academic impact.

Recommendations

These suggestions support student well-being but are not one-size-fits-all.

Educational Institutions

→ Track key digital-wellness signals
→ Use short questionnaires to flag risk
→ Provide simple dashboards for advisors

Student Affairs

→ Run workshops on digital balance & sleep
→ Use behavior patterns to anticipate risk

Advisors & Counselors

→ Add quick digital-wellness check-ins
→ Help students set screen-free study routines
→ Suggest habit tools

Open to any questions!

Thank You!